Pricing a Chokepoint: A Synthetic-Control Estimate of the Brent Crude Premium from the 2026 Strait of Hormuz Disruption

Cherryl Chico, Elvis Casco, Nikoloz Darsalia

Master Thesis, Data Science for Decision Making
Barcelona School of Economics

June 2026

Abstract

Roughly a fifth of the world's petroleum-liquids consumption transits the Strait of Hormuz, so its disruption on 28 February 2026 was widely expected to move global oil prices. Measuring that effect is not straightforward. Because Brent crude is a unique global benchmark, no untreated control unit exists from which a counterfactual price can be inferred directly. We construct that missing counterfactual with the *synthetic control method* (SCM), replacing the geographic donors of conventional applications with a pool of 19 macro-financial series (metals, agriculturals, equities, foreign exchange, rates and credit, and volatility) chosen to share Brent's exposure to common global factors while staying off the oil-supply causal path. To keep the result from hinging on any single modelling choice, the synthetic is the median of an ensemble of five estimators with different inductive biases. We first benchmark the design on the Russian invasion of Ukraine of 24 February 2022, the closest comparable historical oil-supply shock, and confirm that the same machinery recovers the documented price move. Applied to Hormuz over the window to late June 2026, it yields an average Brent premium of roughly 53 % (ensemble median, interquartile range 48 % to 53 %) that is robust to placebo-based inference and a battery of robustness and contamination checks, the latter indicating that any residual bias is conservative. The result is a reduced-form, data-driven complement to the structural scenario estimates of energy institutions, and the paper doubles as a transparent template for validating cross-asset synthetic control under a single-unit geopolitical treatment.

Contents

1 Introduction

Since 28 February 2026, shipping through the Strait of Hormuz has been largely halted amid the 2026 Iran war, the chokepoint sitting at the centre of the conflict (World Bank, 2026). The strait is the narrowest binding constraint in the global oil system, with roughly one-fifth of world petroleum-liquids consumption (U.S. Energy Information Administration, 2025) and about 27% of global seaborne oil trade (Congress.gov, 2026) transiting the passage. What makes the chokepoint so important is the absence of a redundant route. Hormuz is a narrow sea lane between Iran and Oman with no maritime bypass, and the overland pipelines that skirt it carry only a small share of the flow (U.S. Energy Information Administration, 2025), so most Gulf crude has nowhere else to go. Even through recent regional conflict, traffic had never been halted, so the 2026 disruption is the first time the strait has actually closed.

Crude oil is an input to almost every industry and a cost borne across nearly all economies, so a disruption on this scale was always going to move prices. The shock registered in Brent, the benchmark against which most internationally traded crude is settled, which rose sharply once the lane was blocked and passed \$100 a barrel for the first time in four years. This is not without precedent. The 2022 Russian invasion of Ukraine drove Brent from about \$95 to \$130 within two weeks, the closest comparable oil-supply shock of the past decade and the benchmark against which we later validate the design.

The size and persistence of the move are what policy turns on. Emergency strategic-reserve releases are sized against an implied price displacement, and the International Energy Agency (IEA) coordinated the drawdown of some 180 million barrels after the 2022 invasion (International Energy Agency, 2022). The monetary-policy choice to look through an oil shock or to tighten depends on how large and how lasting it is. Fiscal stabilisation in subsidising oil importers scales with the move. Sovereign hedging programmes price disruption scenarios against exactly such magnitudes, the largest being Mexico's annual Hacienda hedge (Calenzani et al., 2023). The 2026 disruption has accordingly drawn rapid analysis from the Kiel Institute, the Oxford Institute for Energy Studies, the World Bank, and the IEA (Kiel Institute for the World Economy, 2026; Oxford Institute for Energy Studies, 2026; World Bank, 2026; International Energy Agency, 2026).

Yet that policy-relevant effect is not the raw price jump. The jump conflates the disruption with everything else moving markets at the same time, so its size is not the disruption's alone, and as a single figure it says nothing about how the effect evolves. We therefore pose a causal question.

> What is the causal effect of the 2026 Strait of Hormuz disruption on the price of Brent crude over a policy-relevant post-event window?

We target the total causal effect rather than any single transmission channel, and we estimate it as a path over the post-event window rather than a single number. Each policy lever above acts on the trajectory of the effect, not on a peak-day figure, so the object we need is the average gap between observed Brent and the price that would have prevailed absent the disruption, traced over the whole window. Recovering that counterfactual path, which has no directly observable analogue, is the methodological core of this paper.

Standard observational designs are ill-suited to it. Brent is a single, globally traded benchmark, so difference-in-differences has no untreated control sharing its trend, and instrumental variables has no credible instrument for a one-off geopolitical event. We use the synthetic control method (SCM) (Abadie et al., 2010; Abadie, 2021), which forms a weighted basket of donor series whose pre-event co-movement with the outcome is high and projects that basket forward as the counterfactual. SCM yields a full counterfactual path, makes its identifying assumptions auditable through the donor pool, and takes no structural stance on whether the move is supply, risk premium, or expectations, since its estimand is a reduced-form total effect. Those institutions reach a price path by the opposite route, imposing an assumed barrel shortfall and propagating it through a structural model. Ours reads the effect from the market and assumes no shortfall, providing a complementary perspective.

This combination, SCM on a crude benchmark with cross-asset donors, has no direct precedent. The closest price application uses cross-country donors for a regulated retail market (Becker et al., 2021), neither of which is available here. Because a single estimate could be an artefact of the donor pool or the modelling choices, we set the Hormuz estimate beside the 2022 Russian invasion, whose price move is already documented. Russia runs on the same machinery but is estimated independently on its own pre-window, and the design recovering its known move is what lends the Hormuz number its credibility. Thus, our contribution is twofold, a data-driven average treatment effect on the treated (ATT) for the Hormuz disruption, and a transparent template for validating cross-asset synthetic control under a single-unit geopolitical treatment.

The remainder of the paper reviews the relevant literature (Section 2), sets out the factor model, estimand, and identifying assumptions (Section 3), describes the data, the estimators, and the validation battery (Section 4), audits the donor pool donor by donor (Section 5), and reports results for both events (Section 6) before concluding (Section 7).

2 Related Literature

Pricing the Hormuz disruption off the observed Brent path sits at the meeting point of three literatures that have largely developed apart. One builds causal counterfactuals for a single aggregate unit, a second measures the oil-price effect of geopolitical shocks, and a third governs inference when a single treated unit is observed over a short post-period.

2.1 Building counterfactuals for a single aggregate unit

The problem of constructing a counterfactual for one treated unit is what synthetic control was invented to solve. The method originates with Abadie and Gardeazabal (2003) and Abadie et al. (2010), who build the counterfactual as a convex combination of untreated donors matched on pre-treatment outcomes and formalise permutation (placebo) inference for the single-unit case. Abadie et al. (2015) extend it to comparative politics, and Abadie (2021) codifies the practical requirements we adopt: a long, shock-free pre-window, donors clean of the treatment (the SCM analogue of Stable Unit Treatment Value Assumption, SUTVA), and a good pre-treatment fit as a precondition for credibility.

The original estimator confines the synthetic to a convex combination of donors and so cannot reach a treated unit outside their hull, a limit later work relaxed in several ways. The augmented SCM of Ben-Michael et al. (2021) adds a ridge bias-correction permitting limited extrapolation, Doudchenko and Imbens (2016) embed SCM, difference-in-differences, and regression in a common regularised-regression family that allows unconstrained weights, and Athey et al. (2021) recast the problem as matrix completion.

Across these methods the weights themselves resist clean interpretation. Convex SCM weights are non-unique when donors are correlated (Abadie and L'Hour, 2021), so many weight vectors yield near-identical fit and no single one should be over-read. We therefore use the whole toolbox, running an ensemble that spans the convex, augmented, and regularised-regression cases, and read cross-model weight dispersion through that non-uniqueness rather than as instability.

Despite this methodological maturity, SCM has rarely been applied to a daily market price. Most peer-reviewed applications estimate effects on non-price aggregates such as GDP, trade, and employment, including under discrete geopolitical shocks. Gharehgozli (2017) estimates the GDP cost of sanctions on Iran and Mancini et al. (2024) the trade effect of the 2022 Russia–Ukraine war.

The closest mechanical precedent to a price application is Becker et al. (2021), who apply SCM to Austrian retail fuel prices with other European countries as donors. However, that design differs from ours in three ways that matter, as its treatment is an endogenous regulation rather than an exogenous shock, its outcome an administered retail price rather than a traded benchmark, and its donors cross-country rather than cross-asset. Juárez et al. (2024) come closer on the treatment side, using an oil-price shock as the trigger, but like Gharehgozli (2017) and Mancini et al. (2024) they measure its effect on regional GDP rather than on a price.

2.2 Measuring the oil-price effect of geopolitical shocks

The methods conventionally used for oil-price effects answer narrower or different questions than ours. The dominant approach is the structural VAR with identified supply, demand, and inventory shocks (Kilian, 2009; Baumeister and Hamilton, 2019), with Hamilton (2009) the narrative anchor for 2007–08. These decompose price changes into structural components but yield no counterfactual path and require identification restrictions for which a chokepoint event offers no standard precedent.

A natural alternative method would regress oil prices on the Geopolitical Risk index of Caldara and Iacoviello (2022), but this recovers the average response to a unit of geopolitical risk pooled across many events, a different object from the effect of one specific disruption, and the index spikes endogenously to the very event we study, so it can serve neither as our estimand nor as a clean donor. Short-window event studies (Brown and Warner, 1985) complete the incumbent set and give a magnitude but not the multi-month path policy requires.

2.3 Inference with one treated unit and a short post-period

Our setting makes inference non-standard, and the recent literature counsels restraint. Permutation and placebo tests (Abadie et al., 2010, 2015) are the field standard, with validity resting on an approximate-symmetry assumption (Canay et al., 2017). Hahn and Shi (2017) show that for convex SCM this effectively requires Gaussian errors, which bounds how far placebo p-values can be pushed for our nonlinear ensemble members, and Chen and Yan (2023) formalise the in-time placebo into a mixed test with a permutation p-value. Because each donor contributes its own placebo p-value, multiple testing across them is controlled by Benjamini–Hochberg (Benjamini and Hochberg, 1995).

The same restraint applies to pretesting. Roth (2022) shows that pre-trend tests have low power at applied sample sizes and can manufacture false confidence, and Rambachan and Roth (2023) propose bounding rather than testing such violations, so we accordingly adopt a report-and-interpret stance toward pre-period diagnostics rather than imposing pass/fail thresholds.

Pre-period fit also bears on donor selection. The value of a donor depends on how reliably it tracks the common factor, and Ham and Miratrix (2024) show that matching on pre-period outcomes before differencing can mask rather than remove contamination when parallel trends fails, a caution we carry into the contamination-bias analysis.

2.4 Where this study sits

Each strand speaks to part of the problem of pricing the Hormuz disruption. Synthetic control has been taken to GDP, trade, and employment but almost never to a daily traded price, and never to a crude benchmark under an exogenous geopolitical disruption with cross-asset donors. The incumbent oil-price methods that might fill that space ask a different question, decomposing structural shocks or pooling geopolitical risk rather than tracing one disruption's counterfactual path. Inference with a single treated unit over a short post-period calls for the restraint this literature counsels. For the Hormuz event the only analysis so far is institutional (Kiel Institute for the World Economy, 2026; Oxford Institute for Energy Studies, 2026; World Bank, 2026; International Energy Agency, 2026; Kilian et al., 2026), each assuming a barrels-per-day shortfall and propagating it through a structural model, none reading the effect off the observed price. This thesis occupies that intersection.

3 Conceptual Framework

This section sets out the data-generating process we assume for Brent and derives the estimand and identifying assumptions from it. The factor model in Section 3.1 is the organising device, treating donor selection as an economic exercise, rendering each identifying assumption transparent, and giving the contamination bias an explicit sign. It also exposes the residual model uncertainty that motivates the ensemble estimator, whose construction we defer to the methodology (Section 4.3).

3.1 A factor model of Brent

We decompose log Brent into a component spanned by global macro-financial factors common to many assets and an oil-specific component:

$$p_t^{\text{Brent}} = \underbrace{\lambda' F_t}_{\text{common global factors}} + \underbrace{o_t}_{\text{oil-specific (supply, chokepoint, geopolitics)}} + \varepsilon_t. \tag{1}$$

F_t collects the five drivers that move Brent in normal times, namely the global industrial cycle, the US dollar, real interest rates, global risk appetite, and China demand, each proxied by non-oil assets. The oil-specific term o_t is the object of interest, the supply and chokepoint component a Hormuz disruption shifts. Each donor j admits its own factor representation $X_{jt} = \gamma_j' F_t + u_{jt}$ with, by construction, zero loading on o_t, so a clean donor is exposed to the same global factors as Brent but is not physically routed through oil supply. A synthetic control is then a weighted donor basket $\hat{p}_t = \sum_j w_j X_{jt}$, and the logic of the method follows directly from (1).

If the weights reproduce Brent's factor loadings, $\sum_j w_j \gamma_j \approx \lambda$, the basket tracks Brent through the pre-window, where o_t is small and stable. When a chokepoint shock fires, o_t jumps for Brent but not for the basket, the two decouple, and that decoupling is the estimated premium. Donor selection is therefore the economic exercise of spanning F_t while excluding o_t, that is, choosing identification over fit. This is precisely why the energy complex, namely West Texas Intermediate (WTI) crude, refined products, and petrocurrencies, despite the highest raw correlation with Brent, is excluded by design. It loads on the very o_t we are trying to measure.

3.2 Potential outcomes and the estimand

Equation (1) describes the observed price, while the causal question concerns the price that would have prevailed without the event. Let Y_t^N be Brent's potential log-price absent the disruption and Y_t^I under it, with treatment beginning at T_0. The treated-unit effect is $\tau_t = Y_t^I - Y_t^N$ for $t \geq T_0$. With only one treated unit, Y_t^N is never observed after T_0. The synthetic supplies the missing term, $\hat{Y}_t^N = \sum_j w_j X_{jt}$, and the estimated effect is the gap $\hat{\tau}_t = Y_t^I - \hat{Y}_t^N$, reported in percent as $\text{gap}_t = 100\,[\exp(Y_t^I - \hat{Y}_t^N) - 1]$.

We summarise the post-window by the *average treatment effect on the treated*,

$$\widehat{\text{ATT}} = \frac{1}{|W|} \sum_{t \in [T_0,\, T_0 + W]} \text{gap}_t. \tag{2}$$

We prefer the ATT to the peak or terminal gap for two reasons. It integrates over the entire path, including any reversion, which is the object that reserve sizing, monetary pass-through, and fiscal buffering act on, and averaging dampens single-day jumps from liquidity or futures-roll mechanics.

3.3 Identifying assumptions

Because the synthetic stands in for an unobserved counterfactual, the estimate is only as credible as the assumptions that license it, and the factor model makes each one explicit and checkable.

1. **Common-factor spanning (pre-period fit).** The donor basket must reproduce Brent's factor loadings, so that good pre-event tracking reflects shared F_t rather than coincidence. This is checked using the pre-period root-mean-square prediction error (RMSPE), walk-forward cross-validation, and moment matching.

2. **Donor cleanliness (SUTVA / no interference).** No donor may carry the treatment, so the loading γ_j must not load on o_t, and donor j must not be hit through a channel specific to the focal event. We used an a-priori audit and confirmed results with model-agnostic break tests at T_0.

3. **Regime stability.** The loadings λ, γ_j must hold across the pre-window and the projection window, so that a basket which tracked Brent before T_0 remains a valid counterfactual after it. This is checked by distance-correlation and structural-break diagnostics, and, across events, by comparing donor weights between regimes.

Assumptions 1 and 3 are testable from observed data, whereas assumption 2 is the genuinely untestable one, since donor cleanliness concerns the unobserved X_{jt}^N. The next subsection shows how the factor model nonetheless lets us sign the direction in which a violation of assumption 2 would move the estimate.

3.4 The direction of contamination bias

The factor representation turns the informal worry that "the donors might be affected by the war" into a quantity with a definite sign. Writing $\delta_{jt} = X_{jt} - X_{jt}^N$ for the event-induced component of donor j, the estimation error is, as an exact identity,

$$\hat{\tau}_t - \tau_t \;=\; Y_t^N - \hat{Y}_t^N \;=\; -\sum_j w_j\, \delta_{jt}. \tag{3}$$

The bias is minus the weight-averaged donor contamination. Its sign is therefore an empirical matter rather than an assumption. If contaminated donors are pushed up (oil-linked terms-of-trade, inflation→rates), the synthetic is pulled up and the ATT is under-estimated, which is conservative. If they are pushed down (demand crowd-out, risk-off), then the ATT is over-estimated.

The identity (3) is exact, but δ_{jt} is itself unobserved, so signing the bias in practice requires estimating each δ_{jt} against a Brent-free reference, a deliberately naive construction whose assumptions and limits we set out, and apply, in Section 6. What the theory delivers here is the structure, which is which way a given contamination pushes the estimate, not a certified magnitude.

The same identity also distinguishes two regimes of contamination by timing. *Contemporaneous* contamination occurs at T_0 and is screened by the audit and break tests. *Delayed* contamination accrues over the post-window, enters through the projection, biases the gap toward zero, and grows with horizon, so a gap that declines as the window lengthens is its signature, and the short-horizon gap is the least-contaminated estimate.

A final source of error is neither identification nor contamination but specification. The counterfactual's functional form is unknown, so no single estimator is privileged. We address this irreducible model uncertainty with an ensemble of five estimators rather than one, a methodological choice we set out with the estimators themselves in Section 4.3.

4 Data and Methodology

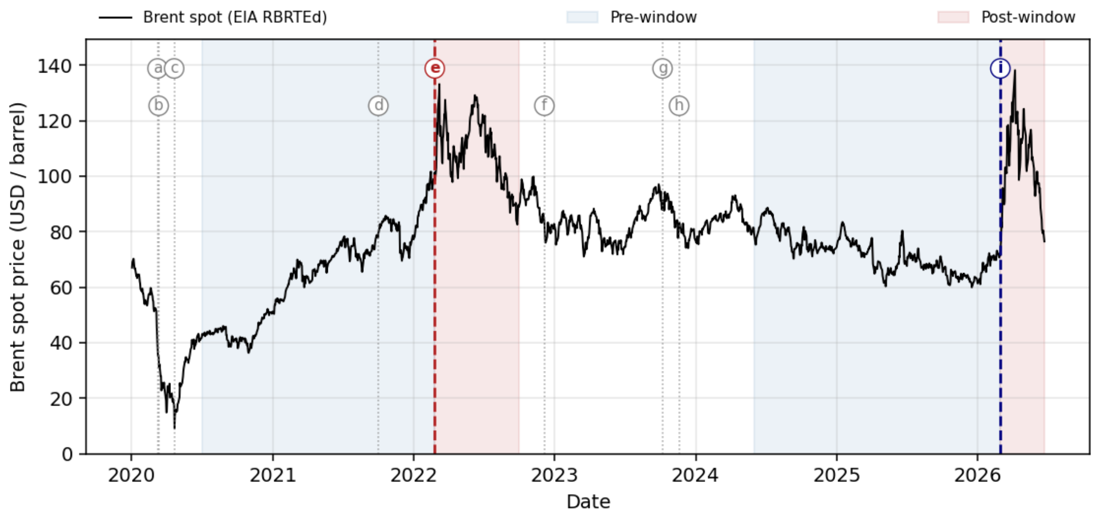
This section sets out the foundations of the empirical design: the data we use, the choices that govern the donor pool and the estimation windows, the estimators that proxy the counterfactual, and the validation battery that defends the estimate. Two themes recur and both follow from the factor model of Section 3. Donor selection is an identification choice rather than a fit-maximising one, and the length of the estimation window is itself a SUTVA consideration, not merely a sample-size one. We describe here what the pool and the methods are and why they were chosen.

4.1 Target and Events

The target is the daily Europe Brent Spot Price FOB (EIA series RBRTEd), quoted in USD per barrel and downloaded directly from the U.S. Energy Information Administration (EIA) (U.S. Energy Information Administration, 2026). The latest available data is as of 22 June 2026, a one-week publication lag. We use Brent because it is the reference price against which most internationally traded crude is settled, and the spot quotation rather than a futures contract so the outcome carries no roll or term-structure mechanics extraneous to the event. The major crude indices are highly correlated, so the choice between them is largely immaterial.

Figure 1 plots the Brent path from 2020 to 2026 against a curated timeline of candidate shocks. Brent roughly doubled after both focal shocks from a similar base near \$75, but reached each along a different trajectory, a steep 2020–22 recovery before Russia and a mild net decline through 2024–25 before Hormuz. Reading the path against the timeline isolates the two events we study, Russia's invasion of Ukraine (24 February 2022) and the Strait of Hormuz crisis (28 February 2026), from the surrounding cluster of lesser shocks.

Figure 1: Brent crude spot price and the curated event timeline, 2020–2026.



Notes. Lettered markers date the curated timeline events. The two focal events (**e**, **i**) are bold dashed lines, the rest grey dotted. (a) Saudi–Russia price war, 8 Mar 2020; (b) COVID-19 pandemic, 11 Mar 2020; (c) WTI-negative, 20 Apr 2020; (d) energy-cycle reflation, 1 Oct 2021; (e) **Russia invades Ukraine, 24 Feb 2022**; (f) Russia oil price cap, 5 Dec 2022; (g) Israel–Hamas war, 7 Oct 2023; (h) Red Sea diversion, 19 Nov 2023; (i) **Strait of Hormuz crisis, 28 Feb 2026**. Shaded bands are each focal event's preferred-specification estimation windows: blue the pre-window (fitting), red the post-window (projection). *Source:* Brent, EIA series RBRTEd. Event timeline curated by the authors (Appendix A.3 for the donor-side audit).

Around each event we choose an estimation window rather than maximise it. A longer pre-history is not automatically better, because Brent and its donors pass through distinct market regimes, and a pre-window spanning several of them teaches the synthetic a donor–Brent relationship that no longer holds in the projection period, a regime shift that re-enters as post-event interference and so as a SUTVA violation. For each event we therefore take the longest pre-window that contains no acute Brent-specific shock or regime break, and we keep the two close to equal in length, about 20 months each, so the cross-event comparison is not confounded by one event training on far more data (Abadie, 2021).

For Russia the pre-window runs from 2020-07-01 to the eve of the invasion. The start is placed deliberately after the March–May 2020 cluster, the Saudi–Russia price war (a), the onset of COVID (b), and the WTI-negative storage failure (c), in which Brent fell about 75 percent peak to trough while the donors fell only 30 to 35 percent. Including that period would teach the synthetic an abnormal Brent factor loading and carry it into the projection. The window does still contain the 2021 energy-cycle reflation (d), but we keep that on purpose. The reflation is a broad and gradual recovery shared across the commodity complex, so it enters through the donors' own co-movement rather than as a Brent-specific break, and the synthetic reproduces it. The Russia post-window then runs from the invasion to 2022-09-30, the last business day before OPEC+'s October 2022 supply cut, a Brent-specific policy shock that would confound the invasion premium. That leaves a post-window of about seven months.

For Hormuz the pre-window runs from 2024-06-01 to the eve of the crisis, about 21 months, and is placed after the 2023 disruptions rather than across them. By mid-2024 the Israel–Hamas war (g) and the Red Sea diversion (h) had been absorbed, the diversion in particular having had roughly six months to transmit through tanker rates and oil-flow patterns, so the window sits inside a single mature post-diversion regime rather than spanning the break. The Hormuz post-window is still open, running from the crisis to the latest available data (2026-06-22), about three and a half months so far.

Table 1: Summary statistics of Brent spot, by event sample.

Event	Segment	Period	n (days)	Mean
Russia 2022	Full sample	2020-07-01–2022-09-30	572	75.95
	Pre-window	2020-07-01–2022-02-23	421	64.26
	Post-window	2022-02-24–2022-09-30	151	108.54
Hormuz 2026	Full sample	2024-06-01–2026-06-22	520	76.98
	Pre-window	2024-06-01–2026-02-27	443	72.08
	Post-window	2026-02-28–2026-06-22	77	105.14

Notes. Prices in USD per barrel. Each post-window begins on the event date T_0, which is the pre/post split.

Table 1 reports summary statistics by event sample. The two pre-windows hold comparable numbers of trading days, 421 for Russia and 443 for Hormuz, so each synthetic is fit on a similar sample. The post-windows are shorter and unequal, 151 days for Russia and 77 for Hormuz, the Hormuz post-window still accumulating as data arrive.

4.2 Donor pool

Donor selection is the central identification choice, and it operationalises the identifying assumptions of Section 3. The candidate pool is 32 macro-financial series from Yahoo Finance that span the asset classes through which the common factors move Brent. A series is admitted only if it meets the two identification conditions of Section 3, common-factor spanning and SUTVA cleanliness, together with three practical filters, daily liquidity from a public redistributable source, enough pre-event history, and an uninterrupted daily series on Yahoo Finance over each event's full estimation window. We fit both events on a single shared pool rather than a separate pool per event, so the basket is held fixed across Russia and Hormuz and the two estimates stay directly comparable. Section 5 implements these conditions as a per-event audit, forms that shared pool, and reports which candidates survive.

4.3 Estimators and the ensemble

A single specification cannot be trusted because the counterfactual's functional form is unknown. Equation (1) need not be linear in the donors, and the convex-hull restriction of classical SCM may or may not bind. Rather than commit to one form, we proxy the counterfactual with five estimators whose inductive biases bracket the plausible cases (Table 2): a simplex-constrained average, a ridge-extrapolating correction, a sparse signed regression, a nonlinear tree ensemble, and a Bayesian linear model. An estimated premium that survives across biases this different is far more likely to reflect a real effect than a single-model artefact. We report the cross-model median as the headline, which is robust to any one model misbehaving, and the interquartile range (IQR) as a model-uncertainty band that complements each model's own within-model interval.

Table 2: The five-estimator ensemble.

Estimator	Inductive bias	Reference
Convex SCM	Non-negative donor weights summing to one, synthetic inside the donor convex hull	Abadie et al. (2010)
Augmented SCM	Convex SCM plus ridge bias-correction, limited extrapolation outside the hull	Ben-Michael et al. (2021)
Elastic net	Signed, sparse+ridge regression, lasso term drops donors entirely	Doudchenko and Imbens (2016)
Gradient boosting (XGBoost)	Nonlinear, tree-based, captures interactions, cannot extrapolate beyond the training range	Chen and Guestrin (2016)
Bayesian ridge	Bayesian linear regression on donor levels (regression component of a Bayesian structural time series, BSTS, model), conjugate Gaussian–Gamma prior	Brodersen et al. (2015)

Notes. Each estimator contributes a distinct inductive bias. The headline is the cross-model median gap across the five.

All five estimators take the same inputs, the log donor levels, and target log Brent. The two events are fit entirely separately. They share the 19-donor pool and the same pipeline, but each trains on its own pre-window, so no parameter learned on one event reaches the other.

We tune hyperparameters inside the pre-window with expanding-window walk-forward cross-validation (Bergmeir and Benítez, 2012; Hyndman and Athanasopoulos, 2018). The first fold trains on the first half of the pre-window and forecasts the next 20 business days, about a month. Each later fold extends the training block to the end of the previous forecast, refits, and forecasts the next 20 days, for five folds in all. We pool the five folds' validation residuals into one out-of-sample root-mean-square error (RMSE) (Hyndman and Athanasopoulos, 2018) and keep the configuration that minimises it. Grids stay narrow, one or two knobs per model over three to nine configurations tuned per event, to limit the search's own overfitting (Cawley and Talbot, 2010). Convex SCM has no tuned hyperparameter.

We then refit that configuration on the full pre-window to build the synthetic and project it onto the post-window. Each model thus implies its own gap series, the difference between observed and synthetic log Brent. At every t the headline is the cross-model median gap across the five, and the interquartile range is the model-uncertainty band. The details of the models and hyperparameters tuning results are in Appendix A.1.

4.4 Validation battery and its logic

A credible SCM estimate must clear three levels (Abadie, 2021), pre-event fit, donor identification, and inference. We run every test per model and per event, itemised by level below.

Fit. One out-of-sample metric.

- **Walk-forward RMSE.** The expanding-window walk-forward cross-validation of Section 4.3 (Hyndman and Athanasopoulos, 2018; Bergmeir and Benítez, 2012) doubles as the fit check. A synthetic whose out-of-sample pre-period tracking matches its in-sample tracking is reproducing a stable donor–Brent relationship rather than memorising the pre-window, which is the empirical content of common-factor spanning. It never selects among the five estimators, all of which are retained.

Identification. This defends the no-interference (SUTVA) assumption and the regime-stability assumption, across donor- and model-level diagnostics.

- **Donor-level cleanliness battery.** Before any model is fit, each donor's own return series is tested for a mean shift at T_0 with a permutation form of the known-break test of Chow (1960) (the date is known, so not the unknown-break test of Andrews (1993)) and a Wilcoxon event-window test (Brown and Warner, 1985), a donor flagged if either rejects under Benjamini–Hochberg false-discovery-rate (FDR) control (Benjamini and Hochberg, 1995). Loading stability is checked with distance correlation across pre-period thirds (Székely et al., 2007), Bai–Perron break dating (Bai and Perron, 1998, 2003), and an I(0)/I(1)-robust trend-break test (Harvey et al., 2009; Kwiatkowski et al., 1992). These operate on the individual donor, so their outcomes are reported with the donor audit (Section 5).
- **Model-level contamination bias.** For the synthetic as a whole, the signed diagnostic of equation (3) gives the direction in which any residual donor contamination would move the estimate, so a surviving gap can be read as conservative or not.
- **Post-window horizon sensitivity.** Re-summarising the gap at 1/2/3/6-month and full horizons traces how any delayed contamination accrues over the projection, since a gap that declines with horizon is its signature (Section 6).
- **Cross-event weight transfer.** Applying each model's Russia weights to Hormuz probes regime stability, the third identifying assumption. We read it as an exploratory probe rather than a pass/fail test (Section 6).

Inference. Several tests on the treated unit.

- **In-space placebo.** Refit each donor as a placebo treated unit and rank Brent's post/pre RMSPE ratio against the resulting distribution (Abadie et al., 2010), reading the permutation p-value with the symmetry caveat of Hahn and Shi (2017).
- **In-time placebo.** Re-run at a fake T_0 six months early, formalised as the mixed placebo test of Chen and Yan (2023).
- **Leave-one-donor-out.** Drop each donor in turn to confirm no single donor drives the gap.

Corroboration and baselines. Two checks sit outside the three levels.

- **External (Russia).** The ensemble-median Russia counterfactual is compared against the EIA's last pre-invasion Short-Term Energy Outlook forecasts, an independent structural counterfactual that shares no information path with the SCM (Section 6).
- **Naive floor.** A deliberately contaminated set of donor-free univariate counterfactuals (random walk, drift, linear trend, and an autoregressive integrated moving average, ARIMA, model selected by the Akaike information criterion, AIC, that reduces to a driftless random walk in both events), a lower bound the donor machinery should beat (Section 6).

Following Roth (2022); Rambachan and Roth (2023), we report and interpret these diagnostics rather than turning them into pass/fail thresholds, and we do not auto-exclude flagged models, because the cross-model IQR is itself the model-uncertainty band. Where a model's pre-period gap series is not flat, we also report a drift-adjusted gap. We fit an ordinary least-squares trend to the pre-period gap, read its slope $\hat{\beta}$ in percent per year, and subtract $\hat{\beta}L$ percentage points for a post-window of length L years, isolating the part of the post-window gap that does not simply continue a pre-event trend. For both events it is small, moving the ensemble-median headline by about a percentage point, so we report both figures.

5 Donor Selection

Donor selection is the identification step of the whole design, because the synthetic can isolate the oil-specific premium only if the pool reproduces Brent through the factors it shares with many assets while carrying none of Brent's own oil-supply exposure. In this section we build that pool by first identifying the common factors that drive Brent and the non-oil series that proxy each. Next, we curate the candidate set against a carefully constructed set of selection criteria. Finally, we confirm with donor-level tests that the qualitative reasoning behind it holds.

5.1 Donor categories and their non-oil proxies

Brent's log price splits into a part driven by global macro-financial factors common to many assets and an oil-specific part (Section 3). A donor is useful when it shares Brent's loading on the common factors but carries no loading on the oil-specific component, because only then does it track Brent before the event and decouple when the oil shock fires. Five factors drive Brent in normal times, and each has a set of non-oil proxies that define the donor categories (Table 3).

Table 3: Common factors driving Brent and their non-oil donor proxies.

Common factor	Economic channel to Brent	Non-oil donor proxies
Global growth / industrial cycle	Oil demand is procyclical, moving with global industrial production and trade	Copper, Iron Ore, broad equities, commodity-exporter FX
US dollar	Oil is dollar-priced, so a stronger dollar lowers its dollar price	the FX block, Gold, DXY (excluded for Russia)
Real rates / monetary policy	Discount rate and carry on a storable commodity, the global liquidity cycle	TLT, US 10-year yield (Mild), Gold
Global risk appetite	Oil is a risk asset and sells off alongside equities and credit when risk turns	VIX (inverse), HYG, S&P 500, safe-haven JPY and CHF, Bitcoin (excluded for Russia)
China demand	China is the marginal buyer of crude and of many commodities	Iron Ore, Copper, Soybeans, CNY, AUD

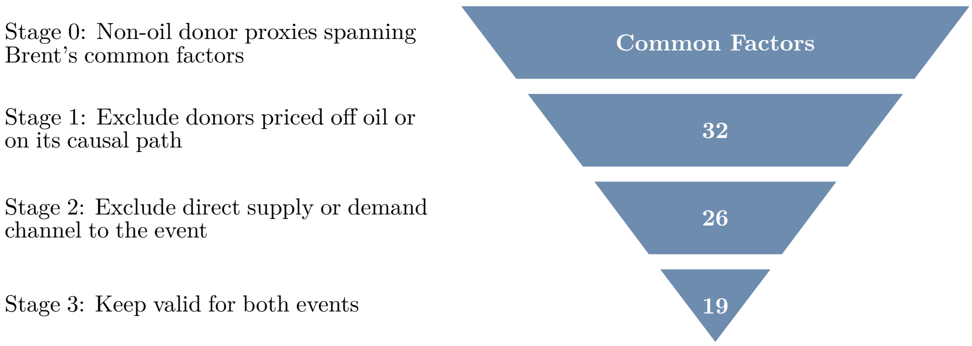
Notes. The pool spans all five factors so the synthetic can reproduce Brent under any mix of drivers, not just one. A series in parentheses is excluded for the event noted.

Industrial and precious metals carry the global growth and real-rate channels through copper, iron ore, gold and silver, whose supply has nothing to do with Gulf or Russian crude. Agriculturals move with global demand and the dollar through the financialised commodity complex, with production concentrated outside the oil chokepoints. Broad equities price global growth and the equity risk premium, with energy only a small index weight. The currency block carries the dollar factor that every non-USD pair shares with dollar-priced oil, together with terms-of-trade and risk channels for commodity exporters and safe havens. Long Treasuries and high-yield credit carry the discount-rate and risk-premium factors, and the VIX carries risk appetite in its purest inverse form. Crypto enters the candidate set through Bitcoin, a digital risk-asset and global-liquidity proxy on that same risk-appetite factor, though the per-event audit excludes it for Russia on the sanctions-evasion channel that opened in March 2022 (Appendix A.3). None of the retained series is itself set by oil.

5.2 Selection criteria

The criterion that binds donor selection is that a donor stay off the path of the disruption we are attributing. Direct supply or demand exposure to the focal event would violate SUTVA and

Figure 2: Donor selection funnel.



Notes. The counts follow the Russia 2022 strict-clean audit, the more demanding event. Stage 2 removes the six donors with direct exposure and Stage 3 the seven with secondary or terms-of-trade exposure, leaving 19 used as the shared pool for both events. Stage 0 is the factor-defined universe and carries no count.

contaminate the very premium we are trying to measure, so we read this criterion strictly in the three stages of the funnel in Figure 2.

Stage 1 removes whole families by construction. The energy complex (WTI, Henry Hub gas, refined products, crude inventories) is priced off the very oil shock we are measuring and would pull the estimated premium toward zero. Petrocurrencies carry that same oil signal in currency form, unstable currencies add domestic-crisis noise rather than shared-factor signal, energy-heavy equity indices carry direct sector exposure, and a few series are dropped as redundant with a cleaner proxy. None of these reaches the per-event step, and Appendix A.3 records each with its reason. What survives is the set of 32 candidates.

Stage 2 drops the candidates with a direct supply or demand channel to the focal event. For Russia 2022 that is six series, the wheat and corn that Russia and Ukraine dominate in world exports, palladium on Russia's share of supply, the euro and the dollar basket through the European energy crisis, and Bitcoin on the sanctions-evasion narrative of early 2022. For Hormuz 2026 nothing is physically routed through the Strait, so none is dropped here. This leaves 26.

Stage 3 keeps only the series clean for both events at once, so the same basket is a valid counterfactual in each. The more exposed event is Russia, so this removes the seven further series that reach Brent through a secondary or terms-of-trade channel over 2022, namely copper, iron ore, soybeans, EM equities, sterling, the US 10-year yield, and cotton. Because the Russia-clean set sits inside the Hormuz-clean set, the survivors stay clean for Hormuz as well. The result is the shared 19-donor pool of Table 4.

Table 4: The retained shared 19-donor pool, by factor category.

Category	Donors (shared pool)
Metals (3)	Silver, Platinum, Gold
Agriculturals (3)	Coffee, Sugar, Live Cattle
Equities (2)	S&P 500, Nikkei 225
Foreign exchange (8)	AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN
Rates & credit (2)	TLT (long Treasuries), HYG (high-yield credit)
Volatility (1)	VIX (implied S&P 500 volatility)

5.3 Empirical confirmation

The donor-level battery of Section 4.3 confirms the audit. For Hormuz 2026, 31 of the 32 candidates agree at the cutoff, and only Cotton differs, its exclusion being an economic causal-channel call that the contemporaneous tests are not expected to reproduce for an indirect, lagged channel. For Russia 2022 the tests add a single flag on CNY, attributable to the concurrent China zero-COVID dynamics rather than to the war. The only within-pre-window relationship break the Bai–Perron procedure dates among retained donors is a single break in the VIX on 2022-01-07, seven weeks before the cutoff (Appendix A.2), which we read as informative rather than exclusionary since it does not recur on the analysis window alone and dropping the VIX moves the estimate by less than one point.

6 Results

6.1 Headline estimate

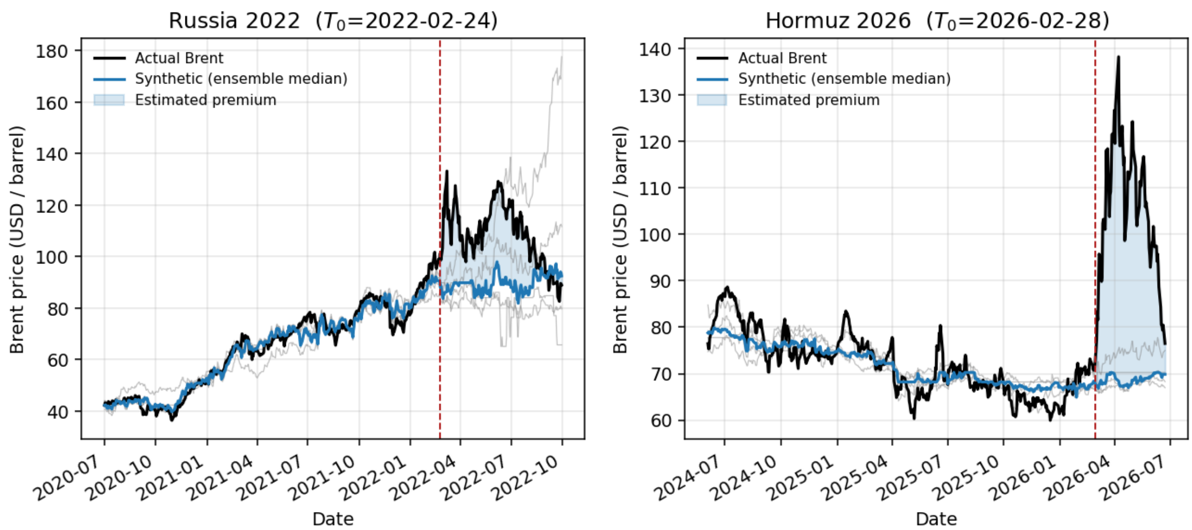
Table 5 reports the per-model post-window gap and the drift-adjusted gap for both events, and Figure 3 plots the observed and synthetic paths. For **Russia 2022** the ensemble-median premium is 22.0 % over the seven-month window (IQR 10.9 % to 30.8 %), which implies a counterfactual mean of \$88.6/bbl against an actual \$108.5, consistent with the documented \$95→\$130 move and its partial reversion by September. For **Hormuz 2026** the ensemble-median premium is 53.1 % over the window to late June (IQR 48.3 % to 53.4 %), with an actual post-window mean of \$105.1 against an implied counterfactual near \$68.7. The Hormuz band is far tighter than Russia's, about 5 points against about 20, which reflects the calmer Hormuz pre-period.

Table 5: Headline post-event gaps (%).

	Russia 2022 (7 mo)		Hormuz 2026 (∼3.8 mo)	
Model	Raw	Adj.	Raw	Adj.
Convex SCM	33.6	28.4	53.1	54.8
Augmented SCM	10.9	10.6	53.7	54.2
Elastic net	22.0	21.2	53.4	54.4
XGBoost	30.8	28.9	48.3	50.2
Bayesian ridge	−9.0	−9.1	40.4	40.4
Ensemble median	**22.0**	**21.2**	**53.1**	**54.2**
IQR	**[10.9, 30.8]**	n/a	**[48.3, 53.4]**	n/a

Notes. Drift-adjusted = raw gap minus the pre-period gap trend extrapolated over the post-window (see Section 4.3). Ensemble = cross-model median, band = IQR (Q25–Q75).

Figure 3: Observed Brent vs. the synthetic counterfactual.



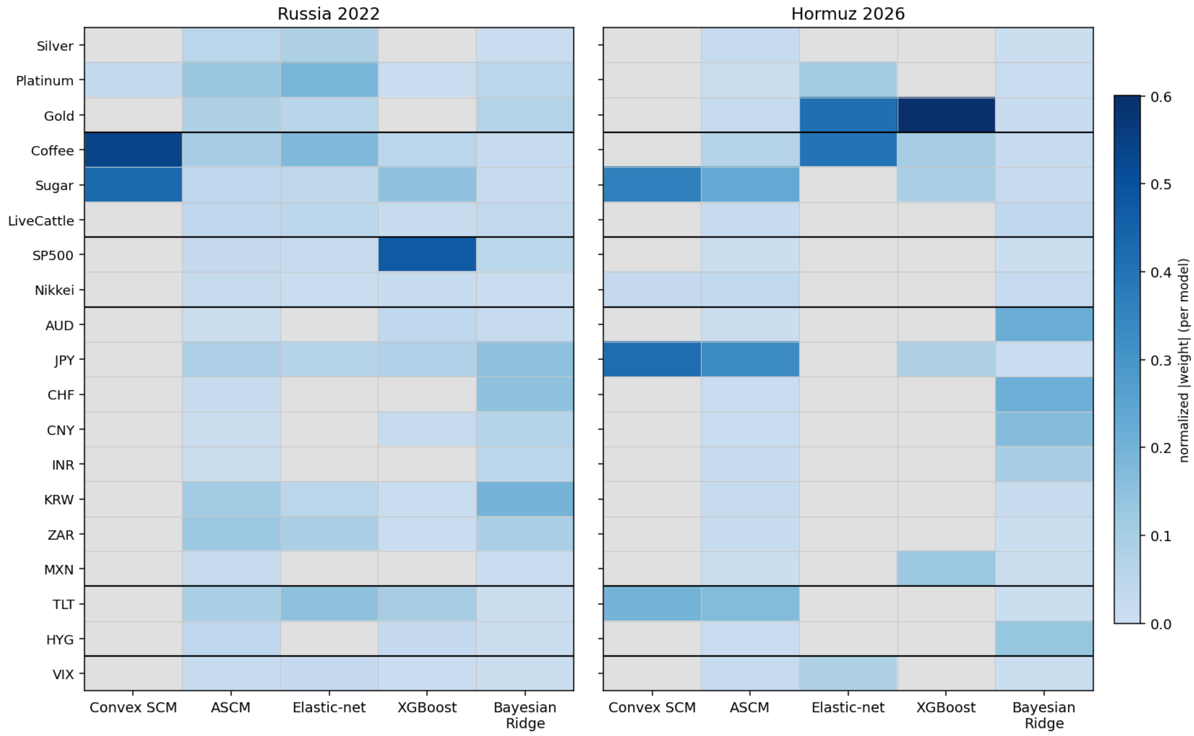
Notes. Observed Brent is the black line. The thick blue line is the ensemble-median synthetic, thin grey lines are the five individual models, and the shaded region after T_0 (dashed red) is the estimated premium.

The cross-model dispersion has an economic explanation rather than reflecting noise, and this is the one place where the individual models matter. Convex SCM, confined to the simplex, leans on the most stable donors, the FX block, as a trend baseline and adds soft-commodity deltas such as Coffee and Sugar for Russia. The Bayesian ridge, with unconstrained Gaussian coefficients, loads heavily and with large opposing signs on currencies. For Russia those currencies moved with the simultaneous and historically large Fed tightening cycle, a macro factor that has nothing to do with the invasion, and that confound is why the Bayesian ridge becomes the Russia outlier, with a negative full-window gap, and why it carries a high walk-forward flag. It is the expected behaviour of an unregularised linear model in a pre-period containing a strong non-event driver, and it is the reason the ensemble uses the median rather than the mean. For Hormuz, whose calmer pre-period has no analogous confound, the five models agree to within about 5 points.

6.2 Donor contributions

To understand why the Russia ensemble is dispersed while Hormuz is comparatively concentrated, we look at the donor contributions behind each model. Figure 4 reports each model's normalised donor weights for both events. The pool is sparse, so most models concentrate on a handful of donors and leave the rest at or near zero, but they disagree on which handful. For example, Convex SCM and XGBoost concentrate weight on two to four donors, augmented SCM and the elastic net spread it across the FX block, and the Bayesian ridge, which shrinks but never zeroes a donor, keeps weight on the whole pool including currencies the others ignore.

Figure 4: Donor weights by model and event.



Notes. Each cell is a donor's weight in one model, taken in absolute value and normalised to sum to one within that model. Darker is larger, grey is an unused (zero-weight) donor. Rows are the 19-donor shared pool grouped by category; columns are the five estimators. The Bayesian ridge is shown for completeness but is excluded from the consensus weight used in the text, as its coefficients are not comparable selection weights (Appendix A.1.1). *Source:* `scripts/donor_contribution_analysis.py`.

How a donor contributes sorts into two descriptive roles, each visible as a row of Figure 5.

- **Trend carriers** carry the largest donor weights, so their combination sets the synthetic's slow level path. Membership here follows the weight rather than a level correlation, since a linear correlation misses donors that contribute through more than straight-line co-movement.

For Russia the carriers (Coffee, Sugar and the S&P 500) climb alongside Brent, while for Hormuz, Brent drifts down through 2024–25 while Gold and Coffee rise, so the synthetic reproduces the decline by loading donors that move the opposite way, and the yen, the second-heaviest donor, stays almost flat and anchors the level.

- **Late divergers** sit on the pool's common factor at T_0 and then peel away over the post-window, shown in the bottom row where each line leaves zero after the cutoff. The role is defined by this behaviour and then attributed to a cause donor by donor, and the row plots only the divergers that also carry weight, since a donor that decouples but is never loaded cannot move the estimate.

  For Russia the weight sits on the Asian currencies, the yen and the Korean won, which the Fed tightening cycle pulled away from the rest of the pool from March 2022; because the Bayesian ridge loads these currencies far more heavily than the other models, with convex SCM putting no weight on them at all, their post-event divergence falls mainly on its synthetic, which is what makes the Bayesian ridge the Russia outlier in Table 5. The renminbi and rupee decouple further still on the discounted-crude terms-of-trade channel audited in Section 5, but the models barely load them, so their divergence is left out of the figure and does not reach the gap. For Hormuz the single weighted diverger is Gold, falling away on its post-strike safe-haven bid.

What separates the models is how much weight they place on a few influential donors and how those donors behave after the event. The Hormuz estimate is driven by the models that lean on Gold, and the Russia estimate by those that lean on the diverging currencies, so the disagreement across models traces back to a handful of heavy donors rather than to noise. This matters for the causal effect because the premium is the gap between actual Brent and the synthetic, and that gap is a clean counterfactual only while the donors hold their pre-event relationship to Brent. Once a loaded donor decouples after T_0, the part of the synthetic resting on it stops standing in for Brent and the donor's own drift leaks into the estimated effect, growing as the window lengthens. For Russia the currencies drift in the conservative direction, lifting the synthetic and trimming the measured premium, so the contamination works against a large effect rather than inventing one. This is why the ensemble takes the median and why the Russia full-window gap is read as the more contaminated of the two estimates. The per-donor diagnostics are in Appendix A.4.

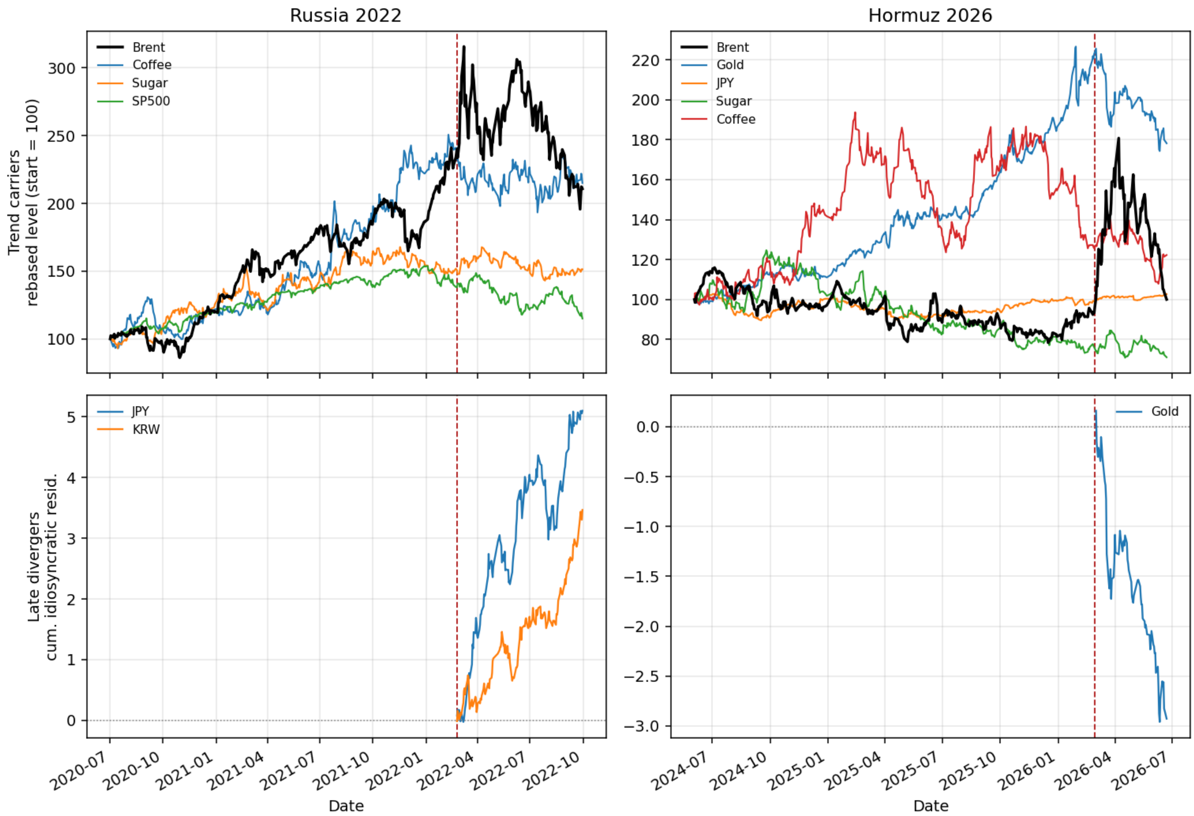
6.3 Validation

A credible estimate must clear three levels: pre-event fit, donor identification, and inference (Section 4.3). We read them in that order.

Fit. Table 6 reports walk-forward cross-validation. The pattern matches the two events' pre-periods. For Hormuz, whose pre-period is calm, four of five models clear the heuristic overfit flag and the fifth sits only just past it. For Russia, whose pre-period spans the volatile 2020–22 recovery, more models trip the flag, because a noisy pre-period inflates validation error. For this analysis, the flags do not exclude any model, we only want to mark which synthetics to read with more caution, and the headline outlier in each event is among the flagged models, so the flags and the outlier tell a consistent story.

Identification. The donor-level cleanliness battery that defends the no-interference (SUTVA) assumption is run before any model is fit and is reported with the donor audit (Section 5). Here we read the three diagnostics that act on the synthetic as a whole: the sign of any residual

Figure 5: Donor paths by contribution role.



Notes. Rows are the two roles and columns are the two events. The dashed red line marks T_0. Trend carriers (top) are the highest-weight donors per event, each rebased to 100 at the pre-window start. For Russia they co-move with Brent in levels, while for Hormuz the heaviest movers (Gold, Coffee) run inversely as Brent drifts down and the yen sits almost flat as a level anchor.
Late divergers (bottom) are the weighted donors that also decouple late, plotted as the standardised cumulative residual of the donor's return on a leave-one-out, Brent-free common factor. Each sits near zero at T_0 and decouples afterward. Donors with a large residual but negligible weight (CNY and INR) are omitted because the models do not load them. The per-donor weights and diagnostics behind both rows are in Appendix A.4.

contamination, how it accrues with horizon, and whether the factor structure is stable enough to carry across regimes.

Sign of the contamination bias. This is a deliberately naive and directional diagnostic, not a structural correction. It does not prove the donors are clean but asks a narrower question. If some contamination remains despite the audit and the break tests, which way would it bias the estimate? We answer it with equation (3) and a Brent-free reference, and the two events give opposite and informative answers. For Hormuz, where every other check already supports a stable estimate, any residual contamination has a small negative sign, between about -1 and -8%. The reading is reassuring in one direction, because if the estimate is biased at all, it is biased downward, so the true premium is at least as large as we report and the headline is conservative. For Russia the sign reverses and the bias is large and positive, on the order of $+13$ to $+21\%$. The reason is economic, because the high-price and risk-off environment of 2022 depressed the demand-cyclical donors, the equities, metals, and soft commodities, relative to the safe-haven and currency donors, so the synthetic built from them sits too low and the gap reads too high. The bias concentrates in the low-reliability soft-commodity donors, and splitting donors by pre-window factor R^2 confirms that most of it sits in the low-reliability group rather than in the donors that track the common factor well. The conclusion is asymmetric, observing that contamination is conservative for the focal Hormuz estimate and inflates the

Table 6: Walk-forward cross-validation (5 expanding folds, 20-day horizons).

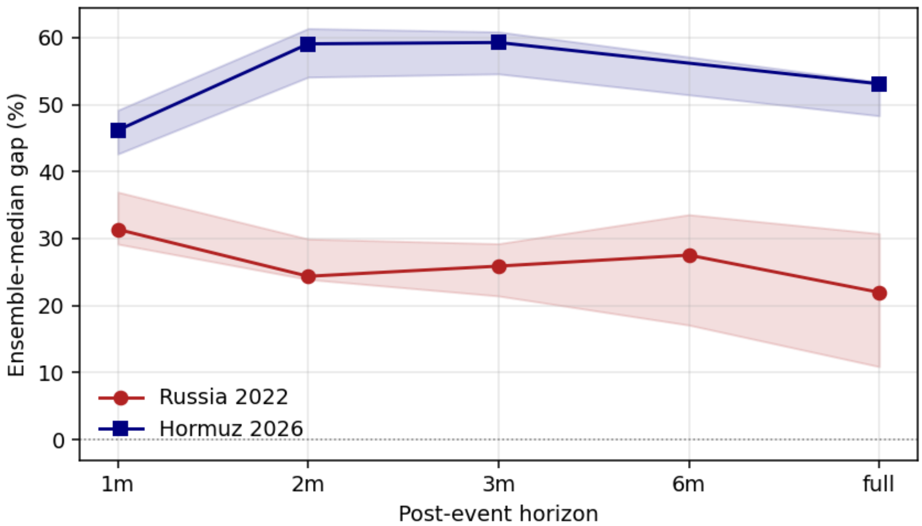
Event	Model	val/train RMSE	Flag
Russia	Convex SCM	1.25	OK
Russia	Augmented SCM	2.06	overfit
Russia	Elastic net	1.99	OK
Russia	XGBoost	3.24	overfit
Russia	Bayesian ridge	3.09	overfit
Hormuz	Convex SCM	1.40	OK
Hormuz	Augmented SCM	1.54	OK
Hormuz	Elastic net	1.51	OK
Hormuz	XGBoost	1.84	OK
Hormuz	Bayesian ridge	2.28	overfit

Notes. A validation/training RMSE ratio above 2 is a heuristic overfit flag, not an exclusion rule.

Russia full-window number, so Russia's magnitude validation rests on the early and peak window, where the \$95→\$130 move is recovered, and not on the attenuated full-window average.

Post-window horizon sensitivity. Figure 6 reports the gap at 1, 2, 3, and 6-month and full horizons. The two events show opposite and diagnostic profiles. Russia declines with horizon, from about 31% to 22%, which is the signature of delayed contamination as second-round war channels switch on through 2022 and pull the synthetic up. The full-window figure is therefore conservative, and the least-contaminated one-month estimate is about 31%. Hormuz rises and then eases, from about 46% to 59% and back to 53%, which is the opposite of attenuation. The chokepoint premium builds in and holds, and the slight easing at the full horizon reflects Brent's partial mid-June pullback that appears once we extend the window to the latest data. The same diagnostic declining for Russia and rising for Hormuz shows that it is not a mechanical artefact.

Figure 6: Ensemble-median gap by post-event horizon.



Notes. The shaded band is the IQR across models. Russia attenuates with horizon, the delayed-contamination signature, while Hormuz rises and then plateaus as the premium builds in and holds.

Cross-event weight stability. As an exploratory probe rather than a validation test, we apply the weights fitted on one event to the other event's window and read how far the pre-period fit degrades. The transfer is weak, and that weakness is informative. Only convex SCM transfers to a clearly positive Hormuz premium, at 23.4% with a degraded but non-pathological pre-fit, while the other models move toward or below zero with pre-fit errors many times their independent level (Table 7). The reason is that the two pre-periods are different factor regimes. Russia's 2020–22 pre-period is shaped by the COVID recovery and the onset of Fed tightening, whereas Hormuz's 2024–26 pre-period is a calmer disinflation regime, so the donor-to-Brent loadings that the weights encode do not carry across.

The models that fail hardest are the ones that load most on currencies, and the dollar and emerging-market FX relationships shifted most between the two regimes. Extending the Hormuz window through the June pullback widens the regime gap further, so we therefore read the transfer as weak and convex-only corroboration that some positive premium survives a change of regime, and as direct evidence that fitting each event separately is the correct choice, not as confirmation of the level.

Table 7: Cross-event weight stability probe: Russia-fitted weights applied to Hormuz.

Model	Indep. gap (%)	Transf. gap (%)	Indep. RMSPE	Transf. RMSPE
Convex SCM	53.1	23.4	0.066	0.336
Augmented SCM	53.7	−2.5	0.066	0.467
Elastic net	53.4	−69.2	0.054	1.513
XGBoost	48.3	−45.2	0.053	0.928
Bayesian ridge	40.4	−100.0	0.037	18.567

Notes. Independent and transferred pre-period RMSPE in log units.

Inference. Table 8 and Figure 7 report the in-space placebo, with the focus on Hormuz. For Hormuz the result is as strong as the test allows. Brent ranks first among all placebo units under every model and both donor pools, with a post/pre RMSPE ratio between 6.7 and 9.9 that exceeds every donor's, so the permutation p-value sits at the attainable floor, 0.050 in the 19-donor pool and 0.029 in the 32-donor pool. For Russia no model reaches the floor, and XGBoost comes closest at $p = 0.20$. This is because the volatile 2020–22 pre-period gives the donors large pre-period RMSPE as well, which widens the placebo distribution and lowers the test's power. This is a known low-power property of the in-space placebo under a volatile pre-period (Abadie et al., 2010), not evidence of a small effect, and we validate Russia's magnitude independently below.

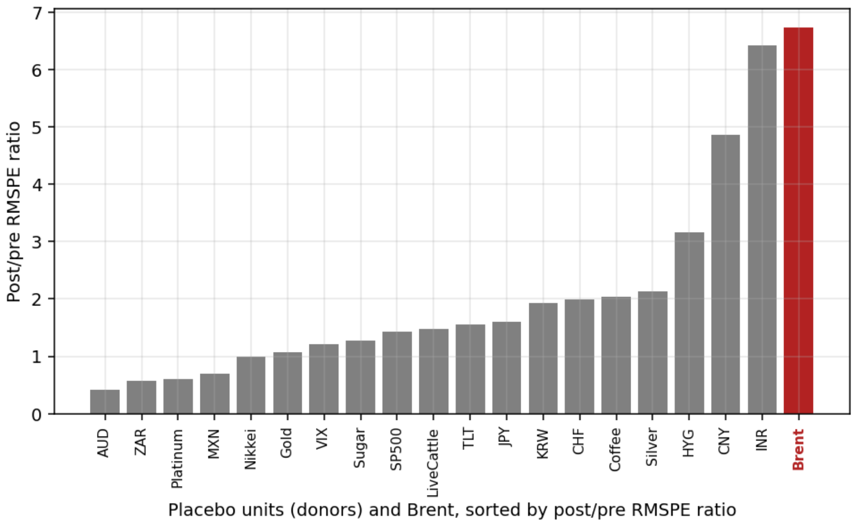
Table 8: In-space placebo: permutation p-values and Brent's post/pre RMSPE ratio.

Event	Model	p (19-donor)	p (32-donor)	Brent ratio
Russia	best of five (XGBoost)	0.20	0.14	7.75
Hormuz	Convex SCM	**0.050**	**0.029**	6.73
Hormuz	Augmented SCM	**0.050**	**0.029**	8.20
Hormuz	Elastic net	**0.050**	**0.029**	8.27
Hormuz	XGBoost	**0.050**	**0.029**	9.89
Hormuz	Bayesian ridge	**0.050**	**0.029**	9.92

Notes. Floor $= 1/(N+1)$: 0.050 (19-donor pool), 0.029 (32-donor pool).

Leave-one-donor-out. Dropping each weighted donor in turn leaves the focal Hormuz estimate tight around its baseline across all models (convex 48.3 to 56.4, augmented SCM 51.6 to 57.4, elastic 52.0 to 58.8, XGBoost 48.8 to 54.8%), so no single donor drives the result. For Russia we report the ensemble-median range rather than every model, because Russia's wider spread comes from the simplex models' reliance on a few soft-commodity donors, the same concentration that drives the contamination finding above. Read that way, the wider Russia range (for example

Figure 7: In-space placebo for Hormuz (convex SCM).



Notes. Each bar is one unit's post/pre RMSPE ratio. Brent (red) is the largest and ranks first among all 19 placebo units, so the permutation test rejects at the attainable floor.

convex 32.8 to 59.6%) is itself a symptom of the donor-quality issue rather than a separate concern.

6.4 Corroboration and baselines

These two checks sit outside the three validation levels: one brings an independent method to bear on the magnitude, the other a donor-free floor the SCM should beat.

Russia magnitude and external validation. Because Russia is a known event, we validate its magnitude against the EIA's last two pre-invasion Short-Term Energy Outlook (STEO) forecasts, issued before 24 February 2022. These forecasts are an independent structural counterfactual that shares no information path with the SCM. Over the identical 151-day window the ensemble-median synthetic mean of \$88.6 sits within \$3.6 of the STEO February 2022 forecast of \$84.9, and three of five models bracket the STEO anchor to within \$4. Two independent methods landing on nearly the same counterfactual is strong evidence that the SCM does not produce an implausible path.

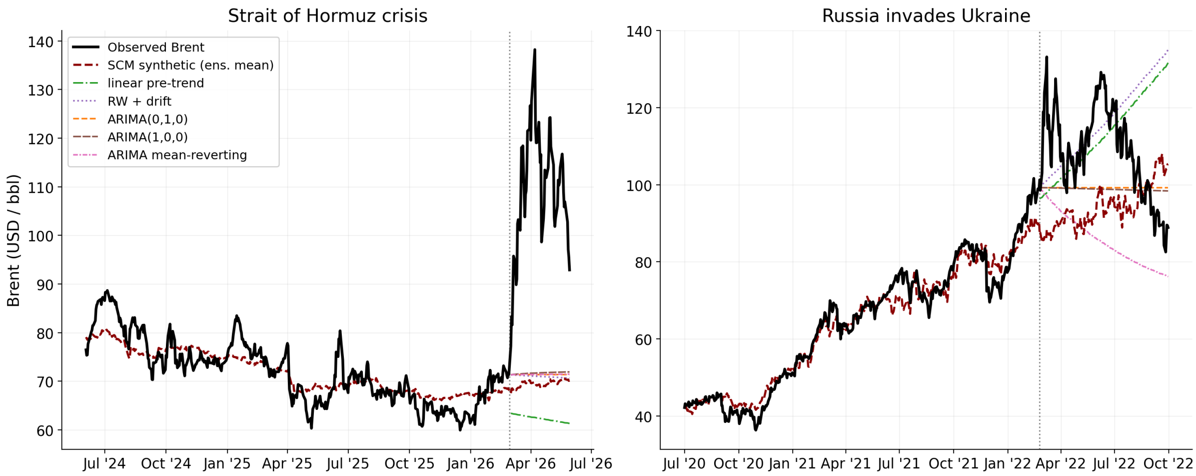
Naive counterfactual floor. We compare the SCM against donor-free univariate counterfactuals built only from Brent's own past (Figure 8), and the comparison yields three findings.

First, the SCM adds value, because it sits about 13 points above the trend-free random-walk null for Russia and about 7 points above it for Hormuz, so the donor machinery extracts something that extrapolating Brent's own series cannot.

Second, and most telling, the naive bias flips sign across the two events purely because the pre-window trend sign differs. The linear trend extrapolates Russia's recovery rally upward and returns an implausible negative full-window gap of about -4%, yet the same method extrapolates Hormuz's mild downtrend and inflates the gap to about 68%. A counterfactual built from one series alone is therefore regime-fragile in a way the donor-based SCM is not.

Third, for Russia the SCM and the trend extrapolation sit on opposite sides of the truth. The SCM leans high, as the contamination analysis above confirms, and the trend extrapolation leans low, so the two bracket the real value, with the random-walk null at +8.7% the most defensible naive point. The naive floor and the contamination analysis thus corroborate each other, one from a donor-free direction and one from a donor-based one.

Figure 8: SCM vs. donor-free naive univariate counterfactuals.



Notes. Observed Brent (black) against the SCM synthetic (red dashed, ensemble) and five counterfactuals built only from Brent's own pre-window: linear pre-trend extrapolation (`lin_trend`), drifted random walk (`rw_drift`), the driftless random walk / ARIMA$(0,1,0)$ carry (`rw_flat`), the best-AIC ARIMA$(1,0,0)$ (`arima_rw`), and an imposed-mean-reversion ARIMA (`arima_mr`).
The dotted line marks T_0; the naive lines are projected over the post-window. Full-window gaps (%), SCM reported as the ensemble median: **Russia** SCM 22.0, `rw_flat` 8.7, `rw_drift` −6.9, `lin_trend` −4.2, `arima_rw` 9.2, `arima_mr` 26.5; **Hormuz** SCM 53.1, `rw_flat` 46.1, `rw_drift` 47.0, `lin_trend` 67.9, `arima_rw` 45.3, `arima_mr` 45.9. The best-AIC ARIMA collapses to the driftless random walk (`rw_flat`) in both events. Generated by `notebooks/08_Milestone3_Plots.ipynb`.

6.5 Interpretation

The 2026 Hormuz chokepoint premium is roughly 53 % of Brent's no-event level over the first nearly four months, and the estimate is stable. It holds across all five estimators and clears the validation battery of Section 4.3, with the in-space placebo reaching its attainable floor as the strongest single result. The contamination analysis adds a useful direction to this stability. If anything, the true Hormuz premium is larger than our estimate, not smaller. The cross-event probe is the weakest of the checks, and we lean on it only for the sign of the effect under a change of regime, never for its level. Taken together, the battery describes a Hormuz premium that is large, stable across estimators, conservatively biased if anything, and driven by no single donor or model.

7 Conclusion

We estimated the Brent-price effect of the 2026 Strait of Hormuz disruption by building a counterfactual with an ensemble synthetic control and benchmarking the design against the documented Russia 2022 shock. Over the window to late June 2026 the ensemble-median premium is roughly 53 % of Brent's no-event level (IQR 48 to 53%), large and positive under every estimator.

The estimate clears the validation battery. In our case, Brent ranks first among all units in the in-space placebo at the attainable floor, the leave-one-out check stays tight, and the contamination-bias sign points conservative. Two further checks corroborate it from outside that battery: the synthetic beats the donor-free naive baselines, and on the benchmark event the same machinery recovers a magnitude consistent with the documented \$95 to \$130 move and within \$3.6 of the EIA's pre-invasion structural forecast, independent reassurance that the method tracks a real path.

The contribution is twofold. The substantive result is a data-driven Brent premium for the Hormuz disruption that complements the structural scenario estimates produced by energy institutions. The methodological result is a transparent template for validating cross-asset synthetic control under a single-unit geopolitical treatment, with an identification argument a reader can audit donor by donor. Our approach and the structural one answer different questions: the structural models ask what price an assumed barrel shortfall implies, while the synthetic control asks what price gap the observed co-movement of global markets implies. Because the two read the event from different directions, they work best together, and our estimate is a complement to scenario analysis rather than a substitute for it.

Several limitations bound the estimate. First, the design has a single treated unit and a short post-period, which caps the power of permutation inference. The Hormuz placebo reaches 0.050, the best value attainable at this sample size rather than a conventional rejection, so we read it as the strongest available signal and not as a small p-value in the usual sense. Second, the estimand is a reduced-form total effect. It captures the full price gap but does not separate the premium into a physical-supply component, a risk-premium component, and an expectations component, which some policy uses would want to distinguish. Third, the Hormuz event is recent and has no documented post-event counterfactual of its own, so its validation is internal and rests on the Russia case as a calibration by analogy rather than on an external ground truth. Fourth, the factor structure is regime-bound, as the weak cross-event transfer shows directly, so the estimate is valid for the regime of this event and should not be read as a general chokepoint premium that would recur unchanged in a different macro environment.

Underlying all four, donor cleanliness is only partly testable. The no-interference assumption concerns an unobserved counterfactual donor path, and the contamination-sign diagnostic is naive and directional, so it bounds the likely direction of any residual bias without proving that none exists. None of these overturns the central finding. The observed co-movement of global markets with Brent implies that the 2026 Hormuz disruption added a large and robust premium to crude, on the order of half its no-event price, and the estimate is meaningful and usable for the decisions that depend on it.

References

Alberto Abadie. Using synthetic controls: Feasibility, data requirements, and methodological aspects. *Journal of Economic Literature*, 59(2):391–425, 2021.

Alberto Abadie and Javier Gardeazabal. The economic costs of conflict: A case study of the Basque country. *American Economic Review*, 93(1):113–132, 2003.

Alberto Abadie and Jérémy L'Hour. A penalized synthetic control estimator for disaggregated data. *Journal of Business & Economic Statistics*, 39(2):492–504, 2021.

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505, 2010.

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Comparative politics and the synthetic control method. *American Journal of Political Science*, 59(2):495–510, 2015.

Donald W. K. Andrews. Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4):821–856, 1993.

Susan Athey, Mohsen Bayati, Nikolay Doudchenko, Guido Imbens, and Khashayar Khosravi. Matrix completion methods for causal panel data models. *Journal of the American Statistical Association*, 116(536):1716–1730, 2021.

Jushan Bai and Pierre Perron. Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1):47–78, 1998.

Jushan Bai and Pierre Perron. Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1):1–22, 2003.

Christiane Baumeister and James D. Hamilton. Structural interpretation of vector autoregressions with incomplete identification: Revisiting the role of oil supply and demand shocks. *American Economic Review*, 109(5):1873–1910, 2019.

Martin Becker, Gregor Pfeifer, and Karsten Schweikert. Price effects of the Austrian fuel price fixing act: A synthetic control study. *Energy Economics*, 97:105207, 2021.

Eli Ben-Michael, Avi Feller, and Jesse Rothstein. The augmented synthetic control method. *Journal of the American Statistical Association*, 116(536):1789–1803, 2021.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57 (1):289–300, 1995.

Christoph Bergmeir and José M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012.

Kay H. Brodersen, Fabian Gallusser, Jim Koehler, Nicolas Remy, and Steven L. Scott. Inferring causal impact using Bayesian structural time-series models. *Annals of Applied Statistics*, 9(1): 247–274, 2015.

Stephen J. Brown and Jerold B. Warner. Using daily stock returns: The case of event studies. *Journal of Financial Economics*, 14(1):3–31, 1985.

Dario Caldara and Matteo Iacoviello. Measuring geopolitical risk. *American Economic Review*, 112(4):1194–1225, 2022.

Jonathan Calenzani, Paul Katz, Sina Sinai, and Klea Kalia. Mexico's petroleum hedging program as counter-cyclical insurance. Research brief, Jain Family Institute, September 2023.

Ivan A. Canay, Joseph P. Romano, and Azeem M. Shaikh. Randomization tests under an approximate symmetry assumption. *Econometrica*, 85(3):1013–1030, 2017.

Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

Qihui Chen and Ge Yan. A mixed placebo test for synthetic control method. *Economics Letters*, 224:111004, 2023.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Gregory C. Chow. Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3):591–605, 1960.

Congress.gov. Iran conflict and the Strait of Hormuz: Impacts on oil, gas, and other commodities. https://www.congress.gov/crs-product/R45281, June 2026. CRS Report R45281, accessed 30 June 2026.

Nikolay Doudchenko and Guido W. Imbens. Balancing, regression, difference-in-differences and synthetic control methods: A synthesis. NBER Working Paper 22791, National Bureau of Economic Research, 2016.

Orkideh Gharehgozli. An estimation of the economic cost of recent sanctions on Iran using the synthetic control method. *Economics Letters*, 157:141–144, 2017.

Jinyong Hahn and Ruoyao Shi. Synthetic control and inference. *Econometrics*, 5(4):52, 2017.

Dae Woong Ham and Luke Miratrix. Benefits and costs of matching prior to a difference in differences analysis when parallel trends does not hold. *Annals of Applied Statistics*, 18(3): 2096–2122, 2024. doi: 10.1214/24-AOAS1872.

James D. Hamilton. Causes and consequences of the oil shock of 2007–08. *Brookings Papers on Economic Activity*, pages 215–261, 2009.

David I. Harvey, Stephen J. Leybourne, and A. M. Robert Taylor. Simple, robust, and powerful tests of the breaking trend hypothesis. *Econometric Theory*, 25(4):995–1029, 2009.

Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, 2nd edition, 2018.

International Energy Agency. IEA member countries agree to new emergency oil stock release in response to market turmoil. News, https://www.iea.org/news/iea-member-countries-agree-to-new-emergency-oil-stock-release-in-response-to-market-turm April 2022.

International Energy Agency. Oil market report (april and may 2026). Technical report, International Energy Agency, 2026.

Luis Juárez, Carlos Chávez, and Mauricio Rojas. When prices go awry: Synthetic control estimates of the regional impact of the 2014 oil-price bust. *Working Paper*, 2024.

Kiel Institute for the World Economy. The cost of closing the strait of Hormuz. Kiel Policy Brief 206, Kiel Institute for the World Economy, 2026.

Lutz Kilian. Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American Economic Review*, 99(3):1053–1069, 2009.

Lutz Kilian, Michael D. Plante, Alexander W. Richter, and Xiaoqing Zhou. The impact of the 2026 Iran war on U.S. inflation: A scenario analysis. Working Paper 2609, Federal Reserve Bank of Dallas, 2026.

Denis Kwiatkowski, Peter C. B. Phillips, Peter Schmidt, and Yongcheol Shin. Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54 (1-3):159–178, 1992.

Michele Mancini, Francesco Paolo Conteduca, and Alessandro Borin. The real effects of the Russia–Ukraine war on trade flows. *World Trade Review*, 23(4):433–447, 2024.

Oxford Institute for Energy Studies. Modelling the Brent price response to a Strait of Hormuz disruption. Energy comment, Oxford Institute for Energy Studies, 2026.

Ashesh Rambachan and Jonathan Roth. A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591, 2023.

Jonathan Roth. Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322, 2022.

Gábor J. Székely, Maria L. Rizzo, and Nail K. Bakirov. Measuring and testing dependence by correlation of distances. *Annals of Statistics*, 35(6):2769–2794, 2007.

U.S. Energy Information Administration. Amid regional conflict, the Strait of Hormuz remains critical oil chokepoint. Today in Energy, https://www.eia.gov/todayinenergy/detail.php?id=65504, June 2025.

U.S. Energy Information Administration. Europe brent spot price fob (series RBRTEd), daily. https://www.eia.gov/dnav/pet/hist/RBRTEd.htm, 2026.

World Bank. Commodity markets outlook, april 2026. Technical report, World Bank, April 2026. http://hdl.handle.net/10986/44610. License: CC BY 3.0 IGO.

Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B*, 67(2):301–320, 2005.

Appendices

A.1 Ensemble models

The ensemble comprises five estimators that map the donor pool to a synthetic Brent series. This appendix states each estimator and reports the hyperparameter grids and the selected settings.

A.1.1 The five estimators

Every member writes synthetic Brent as a weighted combination of the donors, and they differ only in how they constrain or penalise those weights, mostly through two penalties. An L_1 penalty, the sum of the absolute weights, pushes some weights to exactly zero and so selects a sparse subset of donors, while an L_2 penalty, the sum of the squared weights, shrinks every weight smoothly toward zero and steadies the fit when donors move together. We present the members from the simplest linear to the most complex nonlinear, and from the most constrained to the least.

Convex SCM (Abadie et al., 2010). A simplex-constrained weighted average of donors,

$$\hat{y}_t = \sum_{j=1}^{J} w_j \, y_{j,t}, \qquad w_j \geq 0, \quad \sum_j w_j = 1. \tag{4}$$

The weights read like portfolio shares and the synthetic cannot leave the donor envelope. The simplex constraint behaves like an L_1 ball, so the optimiser drives most w_j to exactly zero. The fit is mechanically sparse and leans on the few donors whose convex combination best reproduces pre-period Brent.

Augmented SCM (Ben-Michael et al., 2021). The convex base is kept and a ridge layer, an L_2 correction, is added that extrapolates each donor's contribution beyond the simplex,

$$\hat{y}_t = \underbrace{\sum_j w_j \, y_{j,t}}_{\text{convex base}} + \underbrace{\sum_j \eta_j \, y_{j,t}}_{\text{ridge correction}}, \qquad \eta = (X^\top X + \lambda I)^{-1} X^\top r, \tag{5}$$

where r is the pre-period residual left by the convex base and X the donor matrix. The effective weight is $w_j + \eta_j$; the η_j are unconstrained, which is what permits limited extrapolation outside the donor hull. Large λ shrinks the ridge layer toward zero and recovers convex SCM; small λ gives it more freedom.

Elastic net (Doudchenko and Imbens, 2016; Zou and Hastie, 2005). The elastic net drops the simplex constraint and lets the weights take either sign, controlling them instead with a combined L_1 and L_2 penalty,

$$\hat{y}_t = \beta_0 + \sum_j \beta_j \, y_{j,t}, \qquad \min_\beta \tfrac{1}{2N} \|y - X\beta\|_2^2 \; + \; \alpha \Big[\rho \|\beta\|_1 + \tfrac{1-\rho}{2} \|\beta\|_2^2 \Big]. \tag{6}$$

Here the L_1 term drives some coefficients to exactly zero and the L_2 term shrinks the rest and stabilises the fit under donor collinearity. Here α is the overall penalty strength and ρ the L_1 share, with $\rho = 0$ pure ridge and $\rho = 1$ pure lasso.

Gradient boosting (XGBoost) (Chen and Guestrin, 2016). Gradient boosting leaves the linear family behind and builds the fit as a sum of K shallow regression trees fitted sequentially to the residuals,

$$\hat{y}_t = \sum_{k=1}^{K} f_k(\mathbf{y}_{\cdot,t}), \qquad f_k \in \{\text{trees of depth} \le d\}, \tag{7}$$

with L_2 regularisation on leaf weights and a learning rate η. It is not a closed-form regression. At each split the builder picks the donor and threshold that most reduce the residual. It captures nonlinear interactions but, by construction, cannot extrapolate outside the training range of any donor, which makes it the one member that brackets the extrapolating linear models from below.

Bayesian ridge (Brodersen et al., 2015). The final member returns to linear regression in a Bayesian form, placing a Gaussian prior on the coefficients and Gamma priors on the precisions,

$$y_t = \beta_0 + \sum_j \beta_j\, y_{j,t} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1/\alpha); \qquad \beta_j \mid \lambda \sim \mathcal{N}(0, 1/\lambda), \quad \lambda, \alpha \sim \text{Gamma}(\cdot). \tag{8}$$

The posterior over β is conjugate and closed-form, so $\hat{y}_t$ carries a native posterior standard deviation. Like the L_2 penalty above, the Gaussian prior shrinks coefficients smoothly toward zero but never sets any to exactly zero, so unlike the convex and elastic-net members it performs no selection and retains every donor. This implements only the regression-on-donors term of the BSTS model of Brodersen et al. (2015) and omits its trend, seasonal, and spike-and-slab selection components.

A.1.2 Hyperparameter tuning

We tune each model on the pooled validation RMSE of the expanding-window walk-forward cross-validation described in the body, fitting it separately for each event on that event's own pre-window and refitting on the full pre-window before projection. We keep the grids deliberately narrow and restrict them to the one or two knobs that materially move each fit, since the winner's-curse bias of Cawley and Talbot (2010) grows with the number of configurations tried. Table A.1 reports the grids alongside the configuration each event selected.

The two events land on genuinely different regularisation strengths, and the pattern is consistent across the models. Russia leaves the larger convex residual, so its augmented SCM asks the ridge layer for more freedom through a small λ, whereas Hormuz fits tightly under the convex base and tolerates the heavy shrinkage of $\lambda = 10$. The elastic net keeps $\rho = 0.2$ for both events, so its penalty is four parts L_2 to one part L_1 and it leans on shrinkage and shared weight across correlated donors rather than hard selection, under a light overall penalty that is firmer for Hormuz. XGBoost stays shallow at depth two to three and captures only low-order interactions, but where Russia keeps around thirty lightly penalised trees, Hormuz stops after about ten under a heavier leaf penalty and so does little nonlinear work on its calmer pre-period. The Bayesian ridge selects the same weight precision $\lambda = 1$ for both events, so its coefficient shrinkage is identical and only the assumed noise level moves. These settings produce the per-model gaps reported in the body.

Table A.1: Hyperparameter grids and selected configurations.

Estimator	Hyperparameter	Search grid	Russia	Hormuz
Convex SCM	none tuned[a]	—	—	—
Augmented SCM	ridge λ	$\{0.01, 0.1, 1, 10, 100\}$	0.1	10
Elastic net	α	$\{0.001, 0.01, 0.05, 0.1, 0.5\}$	0.001	0.01
	ρ	$\{0.2, 0.5, 0.8\}$	0.2	0.2
XGBoost[b]	`max_depth`	$\{2, 3, 4\}$	2	3
	`learning_rate`	$\{0.001, 0.005, 0.01, 0.03, 0.1\}$	0.1	0.1
	`reg_lambda`	$\{1, 10\}$	1	10
Bayesian ridge	weight prec. λ	$\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1\}$	1	1
	noise prec. α	$\{10^{-8}, 10^{-6}, 10^{-4}\}$	10^{-4}	10^{-8}

Notes. Grids as run in the fitting pipeline. Selection is by pooled walk-forward validation RMSE, tuned separately for each event on its own pre-window, with the final model refit on the full pre-window.

[a] Convex SCM has no tuned hyperparameter. It draws 120 importance matrices V from a Dirichlet, refits weights for each, and keeps the V with the lowest pre-period RMSPE.

[b] XGBoost sets `n_estimators` $= 1000$ with early stopping after 20 rounds on each fold's validation window and `gamma` $= 0.1$, carrying the median best iteration to the final refit. The selected fits early-stop near 32 trees for Russia and 10 trees for Hormuz.

A.2 Donor-level battery tests

The qualitative audit of Section 5 is the primary identification defence. This appendix reports its empirical counterpart, a battery of donor-level tests run on each candidate's own return or level series before any model is fit. The battery asks two questions. The first is whether a donor jumps at the event, a direct reading of the no-interference assumption. The second is whether the donor's relationship with Brent stays stable across the fitting window, since an unstable loading would distort the synthetic. Every test is model-agnostic and works one donor at a time, so it confirms or supplements the audit rather than replacing it. Throughout we write $r_{j,t}$ for donor j's daily log-return, y_t for Brent's, T_0 for the known cutoff, and $m = 32$ for the number of candidates.

A.2.1 Interference at the event

These tests read each donor's own returns for a shift at T_0, with n_{pre} and n_{post} the counts of pre-cutoff and post-cutoff observations. Two tests carry the flag and a third is reported alongside.

1. **Permutation mean-shift test.** The statistic is the post-minus-pre difference in mean return,
$$d_j = \frac{1}{n_{\text{post}}} \sum_{t>T_0} r_{j,t} \; - \; \frac{1}{n_{\text{pre}}} \sum_{t\le T_0} r_{j,t}. \tag{9}$$
Its null distribution comes from reshuffling the pre and post labels B times and recomputing the statistic on each draw, which gives the p-value
$$p_j = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{\, |d_j^{(b)}| \ge |d_j| \,\}. \tag{10}$$
This is the known-date permutation counterpart of the structural-break test of Andrews (1993) and needs no normality assumption.

2. **Wilcoxon event-window test.** Let E be a short window of returns around T_0 and C a sixty-day pre-window control, with sizes n_E and n_C and pooled size $N = n_E + n_C$. Ranking the pooled returns and summing the ranks that fall in E gives the rank-sum
$$W = \sum_{t\in E} \operatorname{rank}(r_{j,t}), \qquad \mathbb{E}_0[W] = \tfrac{1}{2}\, n_E(N+1), \tag{11}$$
and the two-sided p-value follows from the Mann–Whitney null. Working in ranks makes it robust to the heavy tails of daily returns.

3. **Benjamini–Hochberg correction.** Each test returns one p-value per donor across the m candidates. Ordering them $p_{(1)} \le \cdots \le p_{(m)}$, the procedure rejects every hypothesis up to
$$k^\star = \max\left\{ k : \; p_{(k)} \le \tfrac{k}{m}\, q \right\}, \qquad q = 0.10, \tag{12}$$
which controls the false-discovery rate at q (Benjamini and Hochberg, 1995). A donor is flagged as potentially treated when either the permutation or the Wilcoxon test rejects after this correction.

4. **Kolmogorov–Smirnov distance** (reported, not in the flag rule). The two-sample statistic is the largest gap between the pre and post empirical distributions of the donor's returns,
$$D_j = \sup_x \big|\hat{F}_j^{\text{pre}}(x) - \hat{F}_j^{\text{post}}(x)\big|. \tag{13}$$
At these sample sizes it rejects on any broad macro-regime shift rather than on event-treatment of the individual donor, so we report it but keep it out of the flag rule.

For Hormuz 2026 the battery agrees with the audit on 31 of the 32 candidates, with no flag surviving the correction. The single disagreement is Cotton, which the audit drops on the indirect oil-to-polyester channel that the contemporaneous tests are not built to reproduce. For Russia 2022 the tests miss the audit's directly exposed donors, because at about 415 pre-period observations the war-driven moves are not extreme enough to survive the correction across the candidate set, and they add one flag on CNY that traces to the concurrent China zero-COVID dynamics rather than to the invasion. The Kolmogorov–Smirnov test flags roughly 21 of the 32 Russia donors, far more than the audit, which is the over-rejection on the 2022 regime shift that justifies keeping it out of the flag rule.

A.2.2 Stability of the donor-Brent relationship

Three tests check that the donor-Brent relationship holds across the pre-window.

1. **Distance-correlation change.** The distance correlation (Székely et al., 2007) measures general dependence and is zero only under independence. From the pairwise distances $a_{kl} = |r_{j,k} - r_{j,l}|$ and $b_{kl} = |y_k - y_l|$, double-centred into A_{kl} and B_{kl}, it is

$$\mathrm{dCor}(r_j, y) = \frac{\mathrm{dCov}(r_j, y)}{\sqrt{\mathrm{dVar}(r_j)\ \mathrm{dVar}(y)}}, \qquad \mathrm{dCov}^2(r_j, y) = \frac{1}{n^2}\sum_{k,l} A_{kl}B_{kl}. \tag{14}$$

   Splitting the pre-window into thirds I_1, I_2, I_3 and computing the distance correlation on each, the statistic is the largest change across them, $\max_{a<b} |\,\mathrm{dCor}_a - \mathrm{dCor}_b\,|$. We read it as a narrative flag rather than an exclusion.

2. **Bai–Perron break dating.** Fitting the donor-Brent return relationship $r_{j,t} = \alpha_i + \beta_i y_t + u_t$ with m unknown breaks that partition the sample, the break dates minimise the global segment sum of squares,

$$(\hat{T}_1, \ldots, \hat{T}_m) = \arg\min_{T_1<\cdots<T_m} \sum_{i=1}^{m+1} \sum_{t=T_{i-1}+1}^{T_i} \left(r_{j,t} - \alpha_i - \beta_i y_t\right)^2, \tag{15}$$

   with the number of breaks chosen by the Bayesian information criterion (BIC) (Bai and Perron, 1998, 2003).

3. **Harvey–Leybourne–Taylor trend-break test.** Because the relationship above regresses on Brent, a break dated at the cutoff is ambiguous, since Brent is treated there too. This test instead reads the donor's own log-level trend and is robust to whether the series is stationary or integrated. It combines a statistic t_0 that is optimal under stationarity with a statistic t_1 that is optimal under a unit root,

$$\mathcal{T}_\lambda = (1-\lambda)\, t_0 + \lambda\, t_1, \tag{16}$$

   where the weight λ is driven by a KPSS stationarity statistic so the test keeps the same asymptotic size in either case (Harvey et al., 2009). The KPSS input is

$$\eta_j = \frac{1}{T^2\hat{\omega}^2}\sum_{t=1}^{T} S_t^2, \qquad S_t = \sum_{i=1}^{t} \hat{e}_i, \tag{17}$$

   the scaled partial sums of the residuals $\hat{e}$ from regressing the donor level on a constant and trend, with $\hat{\omega}^2$ a long-run variance estimate (Kwiatkowski et al., 1992). The exact weight function is given in Harvey et al. (2009).

Most donor breaks predate both pre-windows, clustering at the post-crisis normalisation and the 2020 COVID crash, and the COVID break falls just before the Russia pre-window start, which vindicates that start date. Inside the pre-windows the dating flags two Russia donors, Wheat, which the audit already excludes, and the VIX on 2022-01-07 about seven weeks before the invasion, and three Hormuz donors, Silver, Platinum, and EM equities, on a common September 2024 precious-metals move with none near the cutoff. A break inside the pre-window is informative rather than disqualifying, so no donor is removed by this test and the VIX is retained.

For the trend-break test all 32 donors classify as integrated, so it avoids the spurious trend breaks a naive levels regression would find, and no donor's own trend breaks within eight weeks of either cutoff. No retained donor was itself trend-treated by either event.

A.3 Donor audit catalog

Table A.2 is the full per-event catalog underlying the audit of Section 5. Each of the 32 candidates is listed with the common factor it loads on with Brent and its Clean (C), Mild (M), or Heavy (H) status for each event; the final column marks membership in the retained shared 19-donor pool. Status follows the donor catalog maintained alongside the analysis code.

Table A.2: Per-event donor audit catalog.

Donor	Factor it loads on with Brent	RU 2022	HZ 2026	In 19
Metals				
Copper	Global industrial cycle (“Dr Copper”)	M	C	
Iron Ore	China industrial demand	M	C	
Silver	Precious + industrial dual factor	C	C	Yes
Platinum	Auto catalysts / South Africa PGM	C	C	Yes
Palladium	Auto catalysts (Russia ∼40 % supply)	H	C	
Gold	Real rates / safe-haven premium	C	C	Yes
Agriculturals				
Soybeans	Global trade activity, China demand	M	C	
Wheat	Russia+Ukraine ∼30 % world exports	H	C	
Corn	Ukraine ∼13 % world corn exports	H	C	
Coffee	Brazil/Vietnam concentrated, climate	C	C	Yes
Sugar	Brazil/India concentrated	C	C	Yes
Cotton	Oil→polyester substitution (indirect oil)	M	M	
Live Cattle	North-American livestock cycle	C	C	Yes
Equities				
S&P 500	US growth / equity risk premium	C	C	Yes
Nikkei 225	Japan growth / safe-haven equity	C	C	Yes
EM equities	Broad emerging markets (held Russia weight)	M	C	
Foreign exchange				
EUR	Eurozone growth (EU energy crisis)	H	C	
GBP	UK growth (UK gas exposure)	M	C	
AUD	Australia / China-demand commodity FX	C	C	Yes
JPY	Safe-haven FX	C	C	Yes
CHF	Safe-haven FX	C	C	Yes
CNY	China managed currency	C	C	Yes
INR	India growth / EM	C	C	Yes
KRW	Korea manufacturing	C	C	Yes
ZAR	EM commodities (PGMs, gold)	C	C	Yes
MXN	EM / Mexico	C	C	Yes
DXY	Trade-weighted dollar (EUR dominant)	H	C	
Rates & credit				
US 10Y	Real risk-free rate / growth (Europe inflation)	M	C	
TLT	Long-duration US Treasuries	C	C	Yes
HYG	US high-yield credit / risk premium	C	C	Yes
Volatility				
VIX	Implied S&P 500 volatility (inverse)	C	C	Yes
Crypto				
Bitcoin	Digital risk-asset / liquidity (sanctions Mar 2022)	H	C	

Notes. C = Clean (retained for that event); M = Mild (secondary/terms-of-trade channel, excluded from the strict-clean pool); H = Heavy (direct exposure, SUTVA violation, excluded). The shared 19-donor pool is the Russia strict-clean set; “Yes” in the last column marks its members.

A.4 Donor contribution metrics

Table A.3 reports the per-donor diagnostics behind the donor-weights subsection (Section 6). For each donor and event: the consensus weight $\bar{w}$ (mean normalised |weight| across the four comparable models, Bayesian ridge excluded), the across-model dispersion σ_w, the pre-window trend correlation $\rho_{\text{lvl}} = \text{corr}(\log \text{Brent}, \log \text{donor})$, the variance correlation $\rho_{\text{ret}} = \text{corr}(\Delta \log \text{Brent}, \Delta \log \text{donor})$, and the standardised post-window idiosyncratic divergence "div" (cumulative residual of the donor's standardised return on a leave-one-out common factor, Brent-free). All are descriptive. Generated by `scripts/donor_contribution_analysis.py`.

Table A.3: Per-donor contribution diagnostics, by event.

	Russia 2022					Hormuz 2026				
Donor	$\bar{w}$	σ_w	ρ_{lvl}	ρ_{ret}	div	$\bar{w}$	σ_w	ρ_{lvl}	ρ_{ret}	div
Silver	0.03	0.03	−0.05	0.07	−1.57	0.01	0.01	−0.56	0.12	−1.27
Platinum	0.09	0.07	0.48	0.18	−1.39	0.03	0.05	−0.58	0.08	−1.55
Gold	0.03	0.03	−0.54	−0.01	−1.62	0.26	0.26	−0.76	0.14	−2.50
Coffee	0.22	0.19	0.88	0.11	−1.59	0.14	0.15	−0.70	0.01	−0.62
Sugar	0.16	0.16	0.88	0.18	−0.74	0.17	0.14	0.63	0.14	0.43
LiveCattle	0.03	0.02	0.90	0.09	−0.98	0.00	0.01	−0.70	0.11	0.33
SP500	0.13	0.20	0.93	0.28	−2.74	0.00	0.00	−0.63	0.17	0.81
Nikkei	0.01	0.01	0.79	0.12	−0.45	0.01	0.01	−0.46	0.03	0.65
AUD	0.01	0.02	0.21	0.03	−1.43	0.00	0.00	0.06	0.02	0.06
JPY	0.06	0.03	0.85	0.07	5.04	0.21	0.17	0.22	0.03	0.45
CHF	0.00	0.01	0.21	0.06	1.42	0.00	0.01	0.72	0.04	1.06
CNY	0.01	0.01	−0.84	−0.05	6.02	0.00	0.00	0.38	0.03	−0.19
INR	0.00	0.00	0.16	0.01	2.48	0.00	0.01	−0.65	0.06	2.09
KRW	0.04	0.04	0.22	0.09	3.49	0.00	0.01	−0.39	0.11	0.77
ZAR	0.06	0.05	−0.61	0.05	1.98	0.00	0.01	0.37	−0.05	0.19
MXN	0.00	0.01	−0.52	0.01	0.37	0.03	0.05	0.21	−0.02	−0.07
TLT	0.08	0.05	−0.80	−0.24	−2.21	0.09	0.09	0.48	−0.20	−0.69
HYG	0.02	0.02	0.46	0.30	−4.22	0.00	0.00	−0.57	0.14	−0.20
VIX	0.01	0.01	−0.56	−0.28	0.20	0.03	0.03	−0.28	−0.12	−0.61

Notes. $\bar{w}$ and σ_w are the mean and standard deviation of the normalised |weight| across convex SCM, augmented SCM, elastic net, and XGBoost. A large |div| ($\gtrsim 2$) flags a donor that decoupled from the pool's common factor after T_0. Trend correlations ρ_{lvl} are over non-stationary levels and should be read qualitatively.

This table backs the role figure in the main text (Figure 5). Trend carriers are read off the consensus weight $\bar{w}$ rather than the level correlation ρ_{lvl}, since a linear correlation misses donors that carry weight through more than straight-line co-movement, such as the Hormuz yen ($\bar{w} = 0.21$ with ρ_{lvl} only 0.22). Late divergers are read off the div column. The ρ_{ret} column explains why no variance-tracker role appears in that figure. Its largest value is about 0.30 for Russia and 0.17 for Hormuz, too low for any donor to reproduce Brent's daily fluctuation, so the category has no members.